

# Cover Page

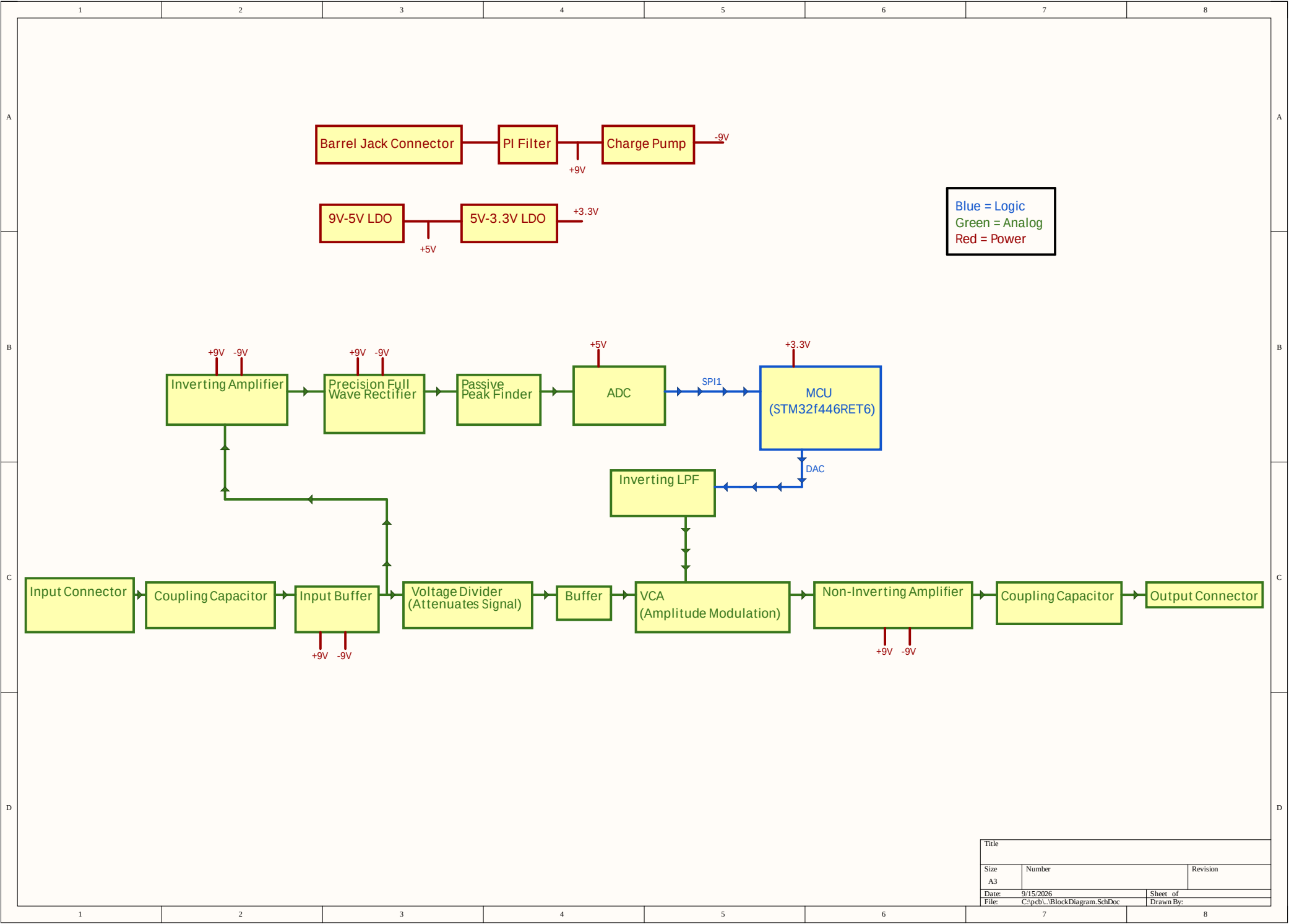
## Page Number

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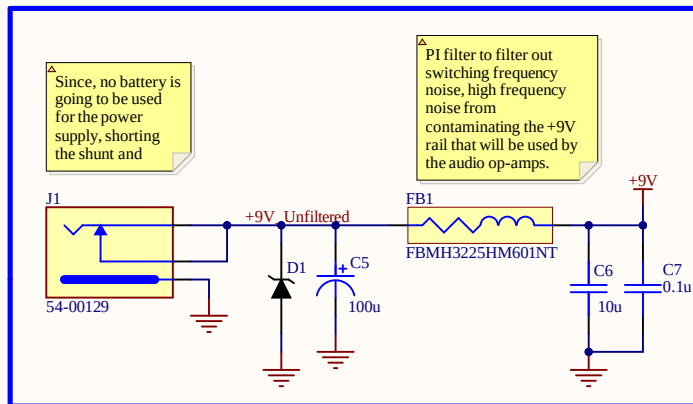
## Description

Cover Page  
Block Diagram  
Power  
MCU  
Analog Signal Path  
Analog Envelope Path  
ADC

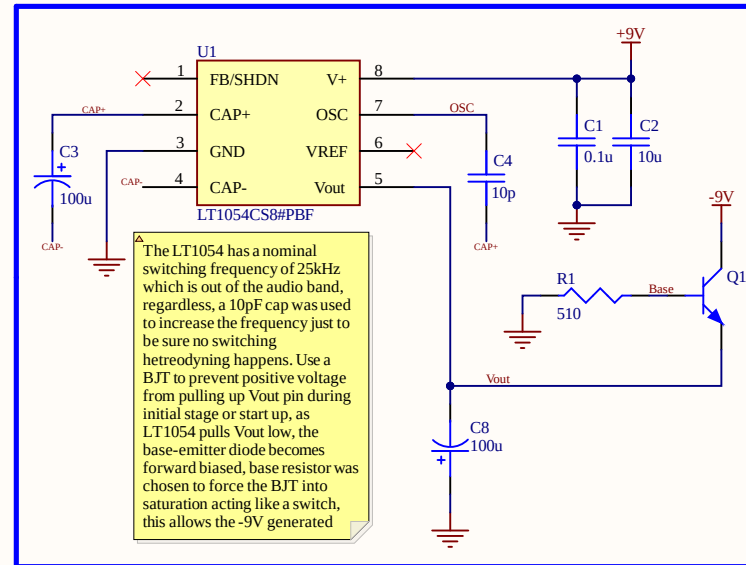
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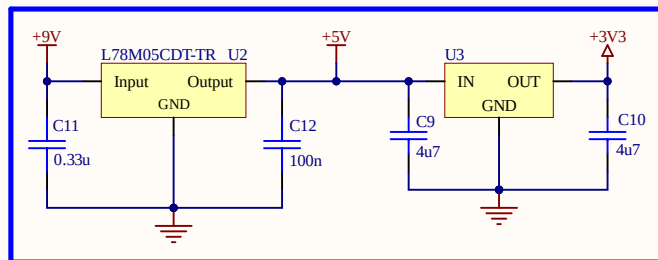
## Barrel Jack Connector and PI filter



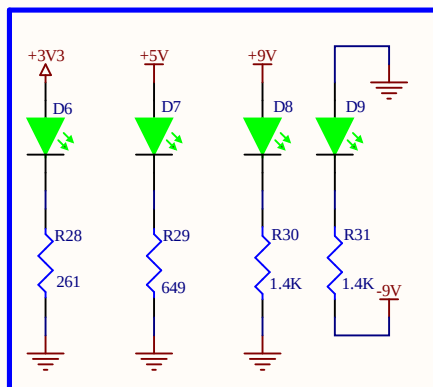
## Charge Pump for Dual Voltage Supply



## LDOs for Clean MCU/ADC Power Rails



## Indicator LEDs



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A circuit diagram showing a parallel combination of six capacitors, labeled C13 through C18. The capacitors are connected between a common top rail and a common bottom rail. The top rail is connected to a +3V3 voltage source, indicated by a red arrow. The bottom rail is connected to ground, indicated by a ground symbol. The capacitors have the following values: C13 = 10u, C14 = 0.1u, C15 = 0.1u, C16 = 0.1u, C17 = 0.1u, and C18 = 0.1u.

Diagram illustrating the oscillator section of the ASDM1-8.000MHZ-LC-T module. The circuit includes a 10nF bypass capacitor (C22) connected to the Vdd pin. The Output pin is connected to the OSC\_IN pin. A callout box states: "10nF bypass cap should be between Vdd and GND pin of the oscillator."

The diagram shows a single-pole single-throw normally open (SPST-NO) switch, labeled SW1. One terminal of the switch is connected to ground. The other terminal is connected to the NRST pin. A capacitor, labeled C23 with a value of 0.1uF, is connected between the NRST pin and ground.

**A** Pull down resistor for BOOT1 pin PB2 allows pin to be used for something else as well as giving use the option to boot from system memory enabling USB flashing. Slide switch allows BOOT0 to be pulled low or high.

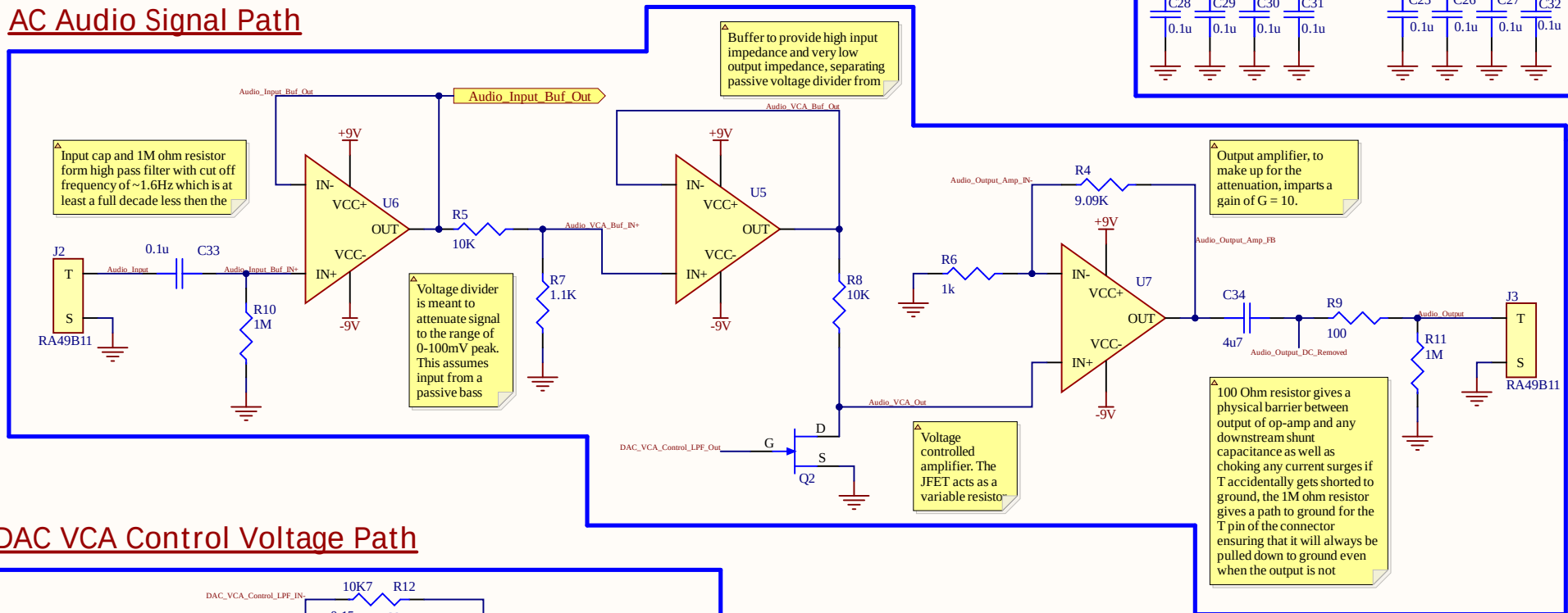
The diagram illustrates the PCB layout for the STM32F446RET6 microcontroller. The chip is represented by a yellow rectangle with pins numbered 1 through 62. The layout includes the following connections and components:

- Power Supply:**
  - VBAT:** Connected to pin 1, with a +3V3 supply.
  - VDD:** Pins 19, 32, 64, and 48 are connected to a common VDD line.
  - VSSA/VREF-:** Connected to pin 12, with a ground connection.
  - VSS:** Pins 47, 63, 18, and 31 are connected to a common VSS line.
  - VDDA/VREF+:** Connected to pin 13, with a ground connection.
- Control Signals:**
  - OSC\_IN:** Connected to pin 5.
  - NRST:** Connected to pin 7.
  - BOOT1:** Connected to pin 28.
  - VCAP\_1:** Connected to pin 30, with a 4u7 capacitor (C24) to ground.
- Peripheral Interfaces:**
  - PC13:** Pin 2.
  - PC14-OSC32\_IN:** Pin 3.
  - PC15-OSC32\_OUT:** Pin 4.
  - PH0-OSC\_IN:** Pin 5.
  - PH1-OSC\_OUT:** Pin 6.
  - PC0:** Pin 8.
  - PC1:** Pin 9.
  - PC2:** Pin 10.
  - PC3:** Pin 11.
  - PA0:** Pin 14.
  - PA1:** Pin 15.
  - PA2:** Pin 16.
  - PA3:** Pin 17.
  - PA4:** Pin 20.
  - PA5:** Pin 21.
  - PA6:** Pin 22.
  - PA7:** Pin 23.
  - PC4:** Pin 24.
  - PC5:** Pin 25.
  - PB0:** Pin 26.
  - PB1:** Pin 27.
  - PB2:** Pin 29.
  - PB10:** Pin 30.
  - PB9:** Pin 62.
  - PB8:** Pin 61.
  - BOOT0:** Pin 60.
  - PB7:** Pin 59.
  - PB6:** Pin 58.
  - PB5:** Pin 57.
  - PB4:** Pin 56.
  - PB3:** Pin 55.
  - PD2:** Pin 54.
  - PC12:** Pin 53.
  - PC11:** Pin 52.
  - PC10:** Pin 51.
  - PA15:** Pin 50.
  - PA14:** Pin 49.
  - PA13:** Pin 46.
  - PA12:** Pin 45.
  - PA11:** Pin 44.
  - PA10:** Pin 43.
  - PA9:** Pin 42.
  - PAB:** Pin 41.
  - PC9:** Pin 40.
  - PC8:** Pin 39.
  - PC7:** Pin 38.
  - PC6:** Pin 37.
  - PB15:** Pin 36.
  - PB14:** Pin 35.
  - PB13:** Pin 34.
  - PB12:** Pin 33.
- Other Connections:**
  - SWCLK:** Connected to pin 49.
  - SWDIO:** Connected to pin 46.
  - SP11\_MOSI:** Connected to pin 57.
  - SP11\_MISO:** Connected to pin 56.
  - SP11\_SCK:** Connected to pin 55.
  - SP11\_CS:** Connected to pin 54.

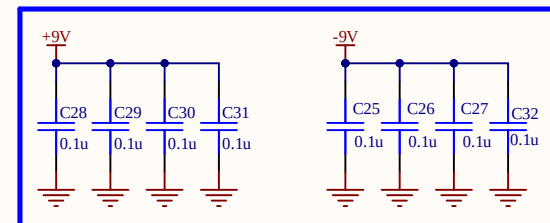
A note indicates: "These traces will need to go on the bottom layer, add stitching vias."

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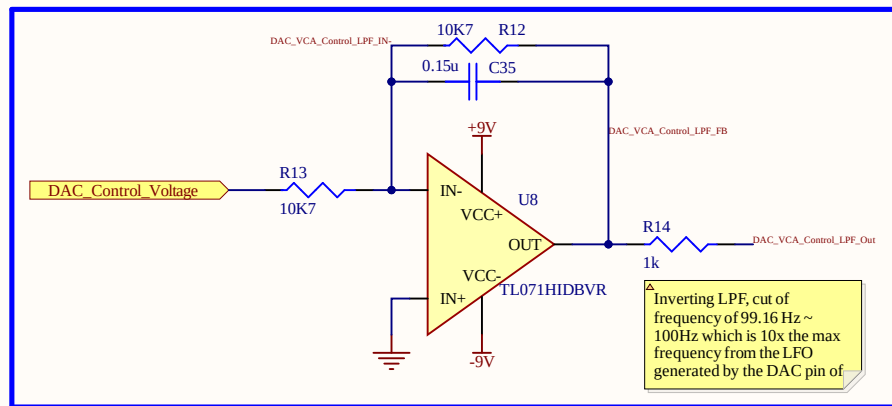
## AC Audio Signal Path



## Coupling Capacitors

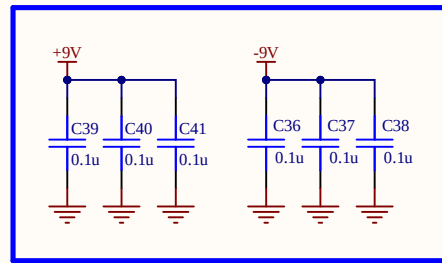


## DAC VCA Control Voltage Path

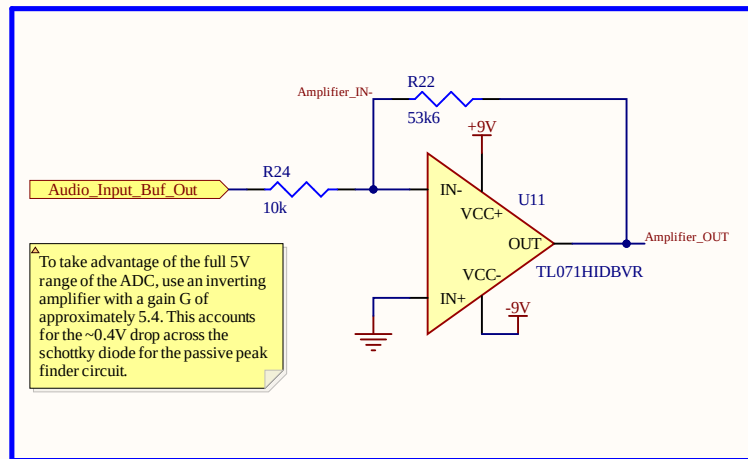


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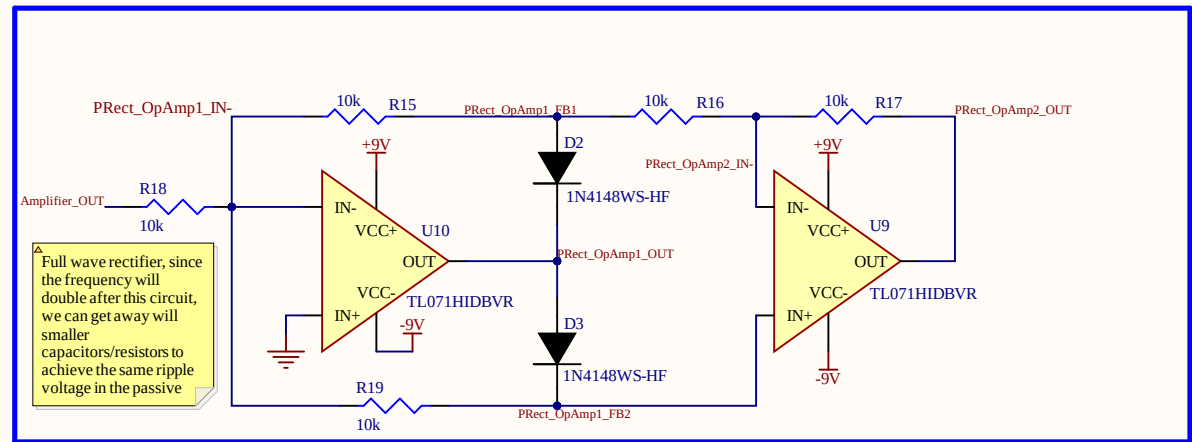
## Decoupling Capacitors



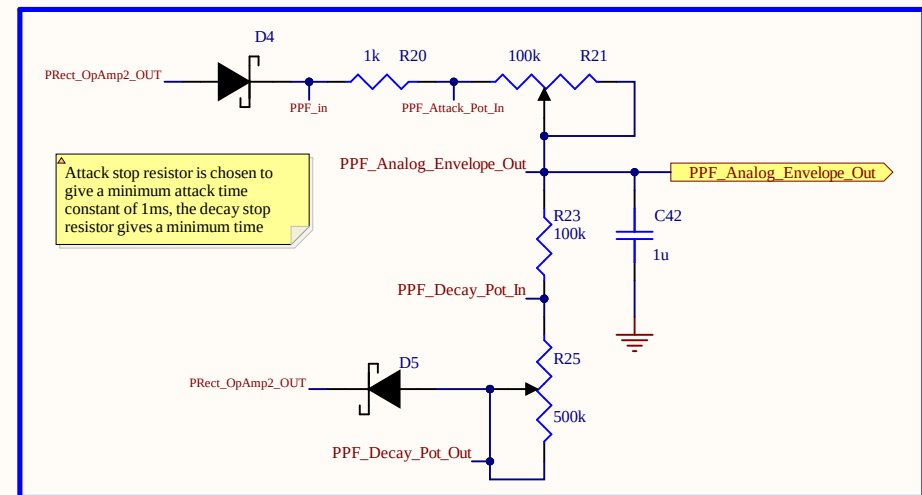
## Inverting Amplifier



## Precision Full Wave Rectifier

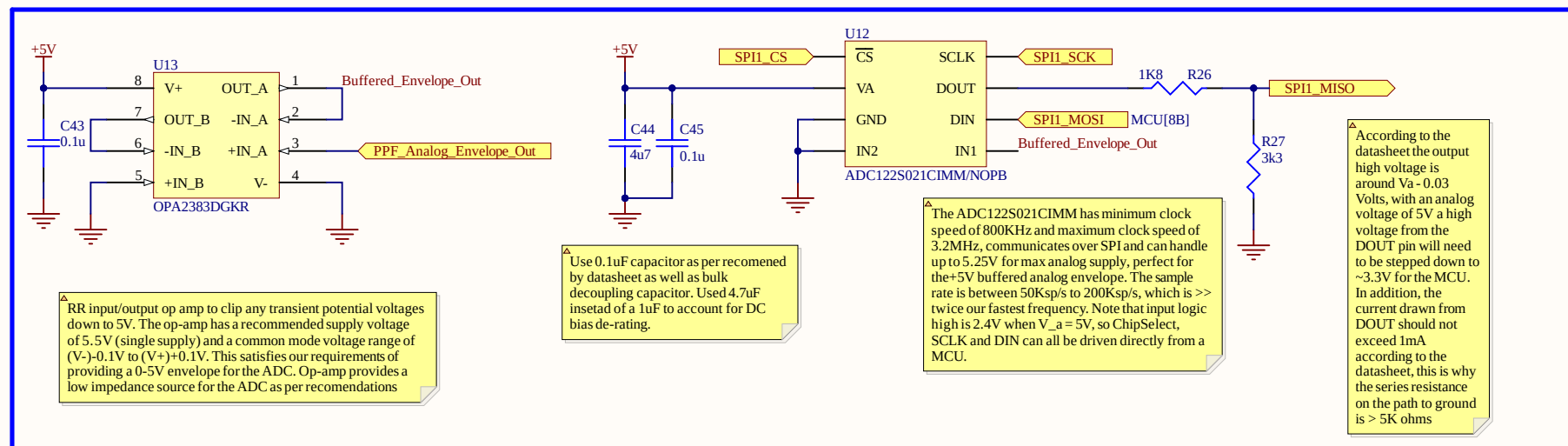


## Passive Peak Finder



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## Input Buffer & ADC



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